

Scheme of Valuation/Answer Key			
(Scheme of evaluation (marks in brackets) and answers of problems/key)			
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY			
SECOND SEMESTER B.TECH DEGREE EXAMINATION(2019 SCHEME), JULY 2021			
Course Code: EST 102			
Course Name: PROGRAMMING IN C (Common to all programs)			
Max. Marks: 100		(AN Session)	Duration: 3 Hours
PART A			
	Answer all Questions. Each question carries 3 Marks		Marks
1	Difference-3 Marks 1. System Software is the type of software which is the interface between application software and system. On other hand Application Software is the type of software which runs as per user request. It runs on the platform which is provide by system software. 2. System software is used for operating computer hardware. On other hand Application software is used by user to perform specific task. 3. Some examples of system software's are compiler, assembler, debugger, driver, etc. Some examples of application software's are word processor, web browser, media player, etc.		(3)
2	Correct algorithm-3 Marks Step 1: start. Step 2: input a, b, c Step 3: let Max=a. Step 4: is b > Max? then Max = b Step 5: is c > max? then Max = c Step6: print Max Step7: stop		(3)
3	Difference-3 Marks The “=” is an assignment operator is used to assign the value on the right to the variable on the left. The '==' operator checks whether the two given operands are equal or not.		(3)
4	static variable-2 marks use- 1 mark Static variables are initialized only once. The compiler persists with the variable till the end of the program. Static variables can be defined inside or outside the		(3)

	function. They are local to the block. The default value of static variables is zero. The static variables are alive till the execution of the program. 1. Static defined local variables do not lose their value between function calls. In other words they are global variables, but scoped to the local function they are defined in. 2. Static global variables are not visible outside of the C file they are defined in.	
5	Correct Program-3 Marks <pre>#include <stdio.h> void main() { char str[100],i; printf("Enter a string: \n"); scanf("%s",str); for(i=0; str[i]!='\0'; ++i); printf("\nLength of input string: %d",i); }</pre>	(3)
6	Array definition-1 mark . Illustration-2 marks	(3)
7	Difference-2 Marks Example-1 mark A structure is a user-defined data type available in C that allows to combining data items of different kinds. Structures are used to represent a record. Each member is assigned unique storage. To define a structure, you must use the keyword struct . A union is a special data type available in C that allows storing different data types in the same memory location. Memory allotted is shared by union members. You can define a union with many members, but only one member can contain a value at any given time. Unions provide an efficient way of using the same memory location for multiple purposes. To define a union, you must use the keyword union	(3)
8	Illustration using example - 3marks A return statement ends the execution of a function, and returns control to the calling function. Execution resumes in the calling function at the point immediately following the call. A return statement can return a value to the calling function. <pre>#include <stdio.h></pre>	(3)

		<pre> void Print() { printf("Welcome to Programming"); } int main() { Print(); return 0; } </pre>	
9		Difference-3 Marks One is an array that refers to an array of characters. The other is a pointer to a character.	(3)
10		fseek()- Syntax-1 mark Use- 2 Marks int fseek(FILE *pointer, long int offset, int position) pointer: pointer to a FILE object that identifies the stream. offset: number of bytes to offset from position position: position from where offset is added. The fseek() function seeks in an open file. This function moves the file pointer from its current position to a new position, forward or backward, specified by the number of bytes.	(3)
PART B			
Answer any one Question from each module. Each question carries 14 Marks			
11	a)	Flowchart- 6 Marks	(6)
	b)	Any 4 functional units.2 marks each	(8)
OR			
12	a)	Registers-1 Mark five important registers in CPU along with functions (5 marks)	(6)
	b)	Algorithm- 4 marks Flowchart -4 marks	(8)
13	a)	Arithmetic operators with example -2.5 marks Logical operators with example - 2.5 marks Bitwise operators with example - 2marks	(7)
	b)	Correct Program-7 Marks	(7)

	<pre> #include <stdio.h> int main() { int i, originalNum, num, lastDigit, sum; long fact; printf("Enter any number to check Strong number: "); scanf("%d", &num); originalNum = num; sum = 0; /* Find sum of factorial of digits */ while(num > 0) { /* Get last digit of num */ lastDigit = num % 10; /* Find factorial of last digit */ fact = 1; for(i=1; i<=lastDigit; i++) { fact = fact * i; } /* Add factorial to sum */ sum = sum + fact; num = num / 10; } /* Check Strong number condition */ if(sum == originalNum) { printf("%d is STRONG NUMBER", originalNum); } else { printf("%d is NOT STRONG NUMBER", originalNum); } return 0; } </pre>	
	OR	

14	a)	Correct Program-7 Marks <pre>#include <stdio.h> #include <math.h> long decimalToBinary(int decimalnum) { long binarynum = 0; int rem, temp = 1; while (decimalnum!=0) { rem = decimalnum%2; decimalnum = decimalnum / 2; binarynum = binarynum + rem*temp; temp = temp * 10; } return binarynum; } int main() { int decimalnum; printf("Enter a Decimal Number: "); scanf("%d", &decimalnum); printf("Equivalent Binary Number is: %ld", decimalToBinary(decimalnum)); return 0; }</pre>	(7)
	b)	Formatted Input – 2marks prototype of 'scanf()' function – 5marks	(7)
15	a)	any 4 string handling functions with syntax and explanation-7 marks strcat(), strrev(), strcpy(),strlen()	(7)
	b)	Correct Program-7 Marks <pre>#include <stdio.h> int main() { int array[100], search, c, n; printf("Enter number of elements in array\n"); scanf("%d", &n); printf("Enter %d integer(s)\n", n); for (c = 0; c < n; c++) scanf("%d", &array[c]); printf("Enter a number to search\n"); scanf("%d", &search); for (c = 0; c < n; c++) { if (array[c] == search) /* If required element is found */ { printf("%d is present at location %d.\n", search, c+1); } } }</pre>	(7)

		<pre> break; } } if (c == n) printf("%d isn't present in the array.\n", search); return 0; } </pre>	
		OR	
16	a)	Correct Program-7 Marks <pre> #include <stdio.h> void main() { int arr1[50], n, i, j = 0, lrg, lrg2nd; printf("\n\nFind the second largest element in an array :\n"); printf("-----\n"); printf("Input the size of array : "); scanf("%d", &n); /* Stored values into the array */ printf("Input %d elements in the array :\n", n); for (i = 0; i < n; i++) { printf("element - %d : ", i); scanf("%d", &arr1[i]); } /* find location of the largest element in the array */ lrg = 0; for (i = 0; i < n; i++) { if (lrg < arr1[i]) { lrg = arr1[i]; j = i; } } /* ignore the largest element and find the 2nd largest element in the array */ lrg2nd = 0; for (i = 0; i < n; i++) { if (i == j) { i++; /* ignoring the largest element */ i--; } } </pre>	(7)

		<pre> else { if(lrg2nd<arr1[i]) { lrg2nd=arr1[i]; } } printf("The Second largest element in the array is : %d \n\n", lrg2nd); } </pre>	
	b)	Correct Program-7 Marks <pre> #include <stdio.h> #include <string.h> void main() { char string[25], reverse_string[25] = {'\0'}; int i, length = 0, flag = 0; printf("Enter a string \n"); gets(string); /* keep going through each character of the string till its end */ for (i = 0; string[i] != '\0'; i++) { length++; } printf("The length of the string '%s' = %d\n", string, length); for (i = length - 1; i >= 0 ; i--) { reverse_string[length - i - 1] = string[i]; } /* Check if the string is a Palindrome */ for (flag = 1, i = 0; i < length ; i++) { if (reverse_string[i] != string[i]) flag = 0; } if (flag == 1) printf ("%s is a palindrome \n", string); else printf ("%s is not a palindrome \n", string); } </pre>	(7)
17	a)	Storage class definition -1 mark. Any 3 storage classes- 1 mark each. Example – 1 Mark each	(7)

	b)	Structure Creation -3Marks Function to read data- 2 Marks Function to display data- 2 Marks.	(7)
		OR	
18	a)	Recursion – 2 marks Correct Program-5 Marks	(7)
	b)	Correct Program-7 Marks	(7)
19	a)	5 file handling functions. Explanation of fopen() fgets() fprintf() fclose() fseek()	(7)
	b)	Correct Program-7 Marks <pre> #include <stdio.h> int main() { char str1[50]; char revstr[50]; char *stpnr = str1; char *rvpnr = revstr; int i=-1; printf("\n\n Pointer : Print a string in reverse order :\n"); printf("-----\n"); printf(" Input a string : "); scanf("%s",str1); while(*stpnr) { stpnr++; i++; } while(i>=0) { stpnr--; *rvpnr = *stpnr; rvpnr++; --i; } *rvpnr='\0'; printf(" Reverse of the string is : %s\n\n",revstr); return 0; } </pre>	(7)
		OR	

20	a)	Differentiate using sample program fragments – 7 marks	(7)
	b)	Correct Program-7 Marks <pre> #include<iostream> using namespace std; #define FILENAME "test.txt" int main() { FILE *fp; char ch; int linesCount=0; //open file in read more fp=fopen(FILENAME,"r"); if(fp==NULL) { printf("File \"%s\" does not exist!!!\n",FILENAME); return -1; } //read character by character and check for new line while((ch=fgetc(fp))!=EOF) { if(ch=='\n') linesCount++; } //close the file fclose(fp); //print number of lines printf("Total number of lines are: %d\n",linesCount); return 0; } </pre>	(7)
NOTE: Marks may be awarded for programs conveying correct logic. Sample programs are included in scheme.			